

Cold Exposure Promotes Atherosclerotic Plaque Growth and Instability via UCP1-Dependent Lipolysis

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Molecular mechanisms underlying the cold-associated high cardiovascular risk remain unknown. Here, we show that the cold-triggered food-intake-independent lipolysis significantly increased plasma levels of small low-density lipoprotein (LDL) remnants, leading to accelerated development of atherosclerotic lesions in mice. In two genetic mouse knockout models (apolipoprotein E(-/-) [ApoE(-/-)] and LDL receptor(-/-) [Ldlr(-/-)] mice), persistent cold exposure stimulated atherosclerotic plaque growth by increasing lipid deposition. Furthermore, marked increase of inflammatory cells and plaque-associated microvessels were detected in the cold-acclimated ApoE(-/-) and Ldlr(-/-) mice, leading to plaque instability. Deletion of uncoupling protein 1 (UCP1), a key mitochondrial protein involved in thermogenesis in brown adipose tissue (BAT), in the ApoE(-/-) strain completely protected mice from the cold-induced atherosclerotic lesions. Cold acclimation markedly reduced plasma levels of adiponectin, and systemic delivery of adiponectin protected ApoE(-/-) mice from plaque development. These findings provide mechanistic insights on low-temperature-associated cardiovascular risks.